

SEASONAL LANDSLIDE INVENTORY FOR NER - 2021



North Eastern Regional Node for Disaster Risk Reduction (NERDRR)

NORTH EASTERN SPACE APPLICATIONS CENTRE

GOVERNMENT OF INDIA, DEPARTMENT OF SPACE,

UMIAM-793103, MEGHALAYA

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12	Abstract	The study was taken up as an ongoing process under NERDRR programme of NESAC, Umiam. It is basically a compendium of all reported landslide incidents collected from newspapers and online media reports. A total of 75 landslide incidents have been documented for the different states of the NER for the year 2021, majority of which have been triggered by rainfall. Further, a total of 21 casualties have been reported out of which 3 are from Arunachal Pradesh, 2 are from Assam, 3 from Manipur, 6 from each Mizoram and Sikkim and 1 is from Tripura. In addition, collapse of road-sections of National/State Highways are reported from all the states of NER with the exception of Tripura. This report also contains information on 24 hrs total rainfall prior to each event. Some of these locations are verified in terms of precise latitude and longitude, either through field survey, photographs and/or post-dated imagery. Accordingly, all codification has been assigned. The systematic inventory thus compiled will help in carrying out future landslide research. Further this will help in assessing the risk in terms of yearly loss of life and property, disruption of transportation network, frequency of landslide and magnitude of landslide incidents in an area and the NER as a whole.		
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PREFACE

North Eastern Regional node for Disaster Risk Reduction (NER-DRR) has been set up at NESAC, Umiam in the year 2011 to provide scientific solutions towards disaster management to the entire region. Almost entire NER consisting of 8 states, namely, Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim and Tripura is susceptible to landslides except the plain areas of Assam, Manipur and Tripura. Physiographically NER is a part of Extra-Peninsula of India and of geologically recent origin, having exposed rocks from Archaean to Quaternary ages. This diverse setup makes the terrain susceptible to landslides. While, intense and/or prolonged monsoonal rainfall and seismic activities are the immediate trigger, anthropogenic factors have a role in accelerating the issue. Hence, the need for immediate and systematic cataloguing of landslide incidents is realized by creating a geospatial database highlighting the landslide locations with respect to nearby settlements and transportation network after the event is reported.

It is expected that, the systematic inventory thus compiled will help in carrying out future landslide research. Further, visualization of the same in spatial context will help in assessing the risk in terms of yearly loss of life and property, disruption of transportation network, frequency and magnitude of landslide incidents in an area and NER as a whole.

DPD, NERDRR

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1. INTRODUCTION

The term "landslide" describes a wide variety of processes that result in the downward and outward movement of slope forming materials composed of natural rock, soils, artificial fills or combinations of these material, (Verns, 1958). According to Geological Survey of India (GSI) reports, 427 k sq. km area accounting 12.6% of the Indian landmass is landslide prone and this contributes app. 8% of global landslide fatalities. Hence, the people living in the North Eastern Region of India are very familiar with both the magnitude and the frequency with which landslides wreak havoc each year. NASA's Landslide Hazard Assessment for Situational Awareness Model (LHASA) classifies Landslide Susceptibility in large parts of North East India under categories 4 and 5 (in a scale from 0 to 5).

Under the purview of North Eastern Regional Node for Disaster Risk Reduction (NERDRR), a need for immediate cataloguing of landslide incidents and creation of a Geospatial database has been realized by highlighting the landslide locations with respect to nearby settlements and transportation network after report of any landslide incident. The systematic inventory thus compiled will help in carrying out future landslide research. Further this will help in assessing State/area wise risk in terms of yearly loss of life and property, disruption of transportation network, frequency of landslide and magnitude of landslide incidents for the whole of NER. This brief report provides information on reported landslide incidences pan NER collected from various sources. However, this may not be inclusive of all the events that have triggered during 2021 monsoon season.

2. SOURCES OF DATA

2.1 LANDSLIDE DATA

Unlike other disasters like earthquakes, so far, landslide incidents are not instrumentally recorded and reported anywhere immediately. Consequently, immediate cataloguing of landslide incidents has been done mainly from newspapers and online/social media. Subsequently, landslide locations are verified in terms of precise latitude and longitude either through field survey* or from post-dated satellite/UAV imagery as and when available. Accordingly, all codification is assigned. The sources of data contributing to this inventory are classified into two categories:

1. Newspapers
2. Online Media

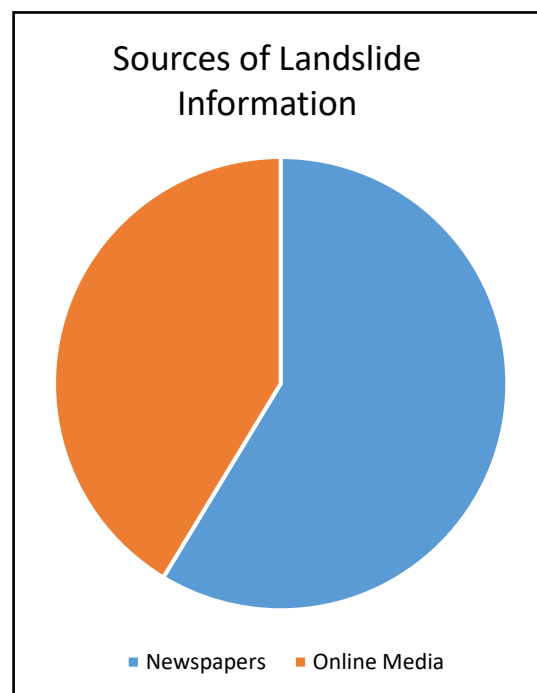


Figure 1: Sources of landslide incidents

Due care is taken to minimise the confusion regarding date of occurrences and precise incident locations of reported events.

Source	Arunachal Pradesh	Assam	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim	Tripura
Total Landslide Events	17	4	11	4	4	15	18	2
Newspapers	10	3	6	2	1	10	11	1
Online Media	7	1	5	2	3	5	7	1

Table 1: Sources of landslide data for different states of NER

* Due to Covid-19 situation, in 2021 field visits are not taken up

2.1.1 NEWSPAPERS:

Newspapers are the primary sources of information that record landslide incidents in the NER. Hence, leading daily newspapers distributed in the different states of NER were thoroughly reviewed either through hard-copies or in the online archives. (Annexure II). These days, online media plays a significant role in reporting disaster incidents. Therefore, data collected from newspapers was supplemented with information acquired from online media.

State wise list of newspapers that are referred for 2021 landslide incidents in NER are provided in Table 2.

Newspaper and Media	Assam	Arunachal Pradesh	Manipur	Meghalaya	Mizoram	Nagaland	Sikkim
The Shillong Times	✓	✓		✓			
The Assam Tribune	✓						
The Sangai Express			✓				
The Sikkim Express							✓
Eastern Mirror Nagaland						✓	
The Morung Express						✓	

Table 2: State wise list of newspapers reporting landslide incidents in NER

2.1.2 ONLINE MEDIA:

In the context of this inventory, online media includes all the small local/regional news stations that only have an online presence. With the ubiquitous accessibility of internet, often landslide incidents photographed/recorded by people end up on different social media platforms. Certain community outreach groups also provide information about recent disasters on such sites. Finally, the army provides updates on twitter whenever it helps in clearing landslides to facilitate movement.

Examples of the same used in this inventory are provided in the Annexure-III.

2.2 SATELLITE DATA

The Satellite imagery available post event or post monsoon are used to extract landslide areas through polygon features to validate positional accuracy of the landslide locations. Freely available imagery from SENTINEL 2 Multispectral Instrument (MSI) is used extraction and updatation of landslide features. Details of the Level 1C and the Level 2A products of SENTINEL 2 MSI used for this study is given below.

The [Level-1C](#) product is composed of 100x100 km² tiles (ortho-images in UTM/WGS84 projection). The Level-1C product results from using a [Digital Elevation Model \(DEM\)](#) to project the image in cartographic geometry. Per-pixel radiometric measurements are provided in Top Of Atmosphere (TOA) reflectances along with the parameters to transform them into radiances. Level-1C products are resampled with a constant Ground Sampling Distance (GSD) of 10, 20 and 60 m depending on the native resolution of the different spectral bands. In Level-1C products, pixel coordinates refer to the upper left corner of the pixel.

Level-1C products will additionally include [Cloud Masks](#) and [ECMWF](#) data (total column of ozone, total column of water vapour and mean sea level pressure).

The [Level-2A](#) product provides Bottom Of Atmosphere (BOA) reflectance images derived from the associated Level-1C products. Each Level-2A product is composed of 100x100 km² tiles in cartographic geometry (UTM/WGS84 projection).

Level-2A products are systematically generated at the ground segment over Europe since March 2018, and the production was extended to global in December 2018.

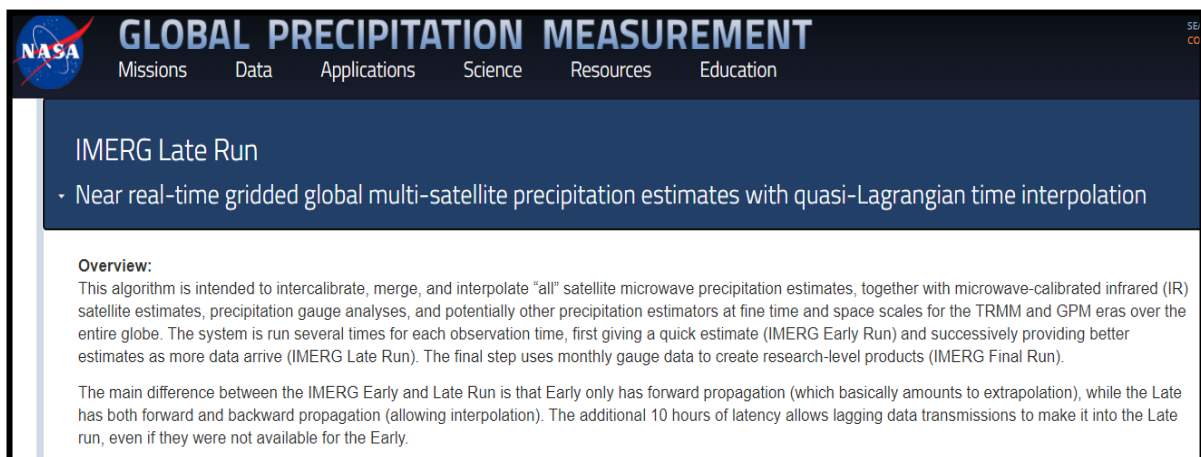
Source: ESA Website

Figure 2: Sentinel 1A and 2C

Both Level 1C and 2A products were used with the Ground Sampling Distance (GSD) of 10m.

2.3 RAINFALL DATA

Rainfall data from NASA's Global Precipitation Mission (GPM) has been used, considering the seamless availability of rainfall data for the entire NER. 'IMERG LATE RUN' product of spatial resolution - 0.1 degrees/~10km has been used.



The screenshot shows the NASA Global Precipitation Measurement website. The header includes the NASA logo and navigation links: Missions, Data, Applications, Science, Resources, and Education. The main heading is "IMERG Late Run", followed by a bullet point: "Near real-time gridded global multi-satellite precipitation estimates with quasi-Lagrangian time interpolation". Below this is an "Overview" section. The overview text states: "This algorithm is intended to intercalibrate, merge, and interpolate 'all' satellite microwave precipitation estimates, together with microwave-calibrated infrared (IR) satellite estimates, precipitation gauge analyses, and potentially other precipitation estimators at fine time and space scales for the TRMM and GPM eras over the entire globe. The system is run several times for each observation time, first giving a quick estimate (IMERG Early Run) and successively providing better estimates as more data arrive (IMERG Late Run). The final step uses monthly gauge data to create research-level products (IMERG Final Run).". A final paragraph explains the difference between the Early and Late runs: "The main difference between the IMERG Early and Late Run is that Early only has forward propagation (which basically amounts to extrapolation), while the Late has both forward and backward propagation (allowing interpolation). The additional 10 hours of latency allows lagging data transmissions to make it into the Late run, even if they were not available for the Early."

Figure 3: GPM IMERG Late Run Product

3. CHARACTERISTICS OF THE LANDSLIDE INVENTORY

The inventory is seasonal in nature and contains events triggered by events of 2021 monsoonal rainfall.

A total of 75 landslide incidents have been documented in the different states of NER. 21 casualties are reported out of which 3 are from Arunachal Pradesh, 2 from Assam, 3 from Manipur, 6 from Mizoram, 6 from Sikkim and 1 from Tripura.

In addition, collapses of road-sections along National/State Highways have been reported from all the states of the NER with the exception of Tripura.

Base and other ancillary information were used to correlate the inventory database is of 1:50,000 scale.

This database reflects the landslide locations immediately after a landslide event is reported in newspaper, online/offline media, preferably within 24hrs of reporting, so depending upon the availability of precise location information the inventory database is grouped into following 2 categories: -

- **TYPE 1 (Represented by Green colour)**

These are events for which precise latitude/longitude information is available. Such events are generally validated from UAV/Satellite Imagery and/or Field Visits. Post-dated satellite imageries are used for validation purpose only where spatial resolution of the image and size and exposure condition of the landslide makes it possible.

- **TYPE 2 (Represented by Yellow colour)**

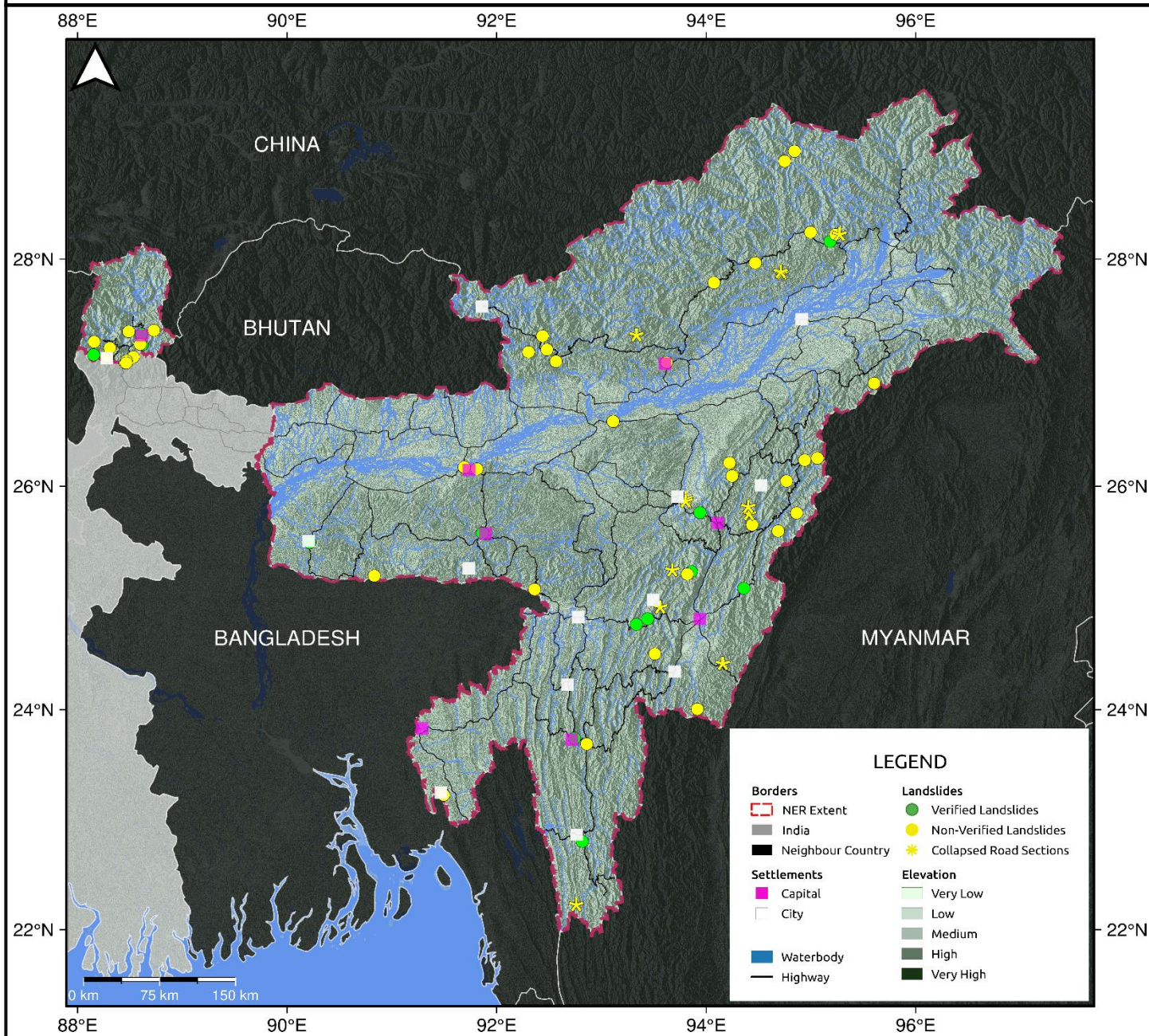
These are events for which precise latitude/longitude information not available rather is referred with respect to nearby village/settlement or locality name by newspapers/media reports. In most of the case these slides are of smaller dimension and could not be visualized using freely available Sentinel 2 data of 10m spatial resolution. However, as and when needed the same may be validated by using post-dated high-resolution satellite data.

- A 3rd category (Type 3) includes areas where effect of landslide on road sections along National Highways/State Highways is reported, which has affected transportation.

Separate symbology is used to distinguish these locations. However, challenge associated for validation of these locations is that, road sections usually get restored or repaired at the earliest possible time thus only a small window is left for validation purpose either through a field visit or using satellite imagery. Revisit status of particular satellite further complicates the situation. However, as and when needed the same may be validated by using post-dated satellite data of appropriate resolution.

All the landslide locations also have standard attribute information, in terms of unique ID, location, date/time/type of events, casualty information, 24hrs rainfall prior to the event etc. (Annexure IV). Due care is being taken to validate and convert all the event locations to Type 1 at the earliest.

Landslide Locations 2021, North East Region, India



Map 1: Landslide Incidents in the NER, 2021

4. METHODOLOGY

4.1 LANDSLIDE POINT AND POLYGON EXTRACTION

For preparation of Seasonal Landslide Inventory post monsoon for the year 2021, a thorough review of articles relevant to landslide incidents from all the major Daily Newspapers in NER was carried out.

The location of all landslide incidents was derived from newspapers and an initial database of landslide locations was prepared in the form of a Point Shapefile.

Of all the landslide incidents, all those events which were determined large enough to be visualized on Sentinel 2 imagery were extracted and a 10km buffer was created around them (for ambiguity resolution in geolocation).

Finally using Pre and Post Event Satellite Imagery, landslide polygons within the 10km buffer wherever visible, were digitized.

The flowchart for the methodology is displayed in Figure 4 and the relevant code is provided in the Annexure V.

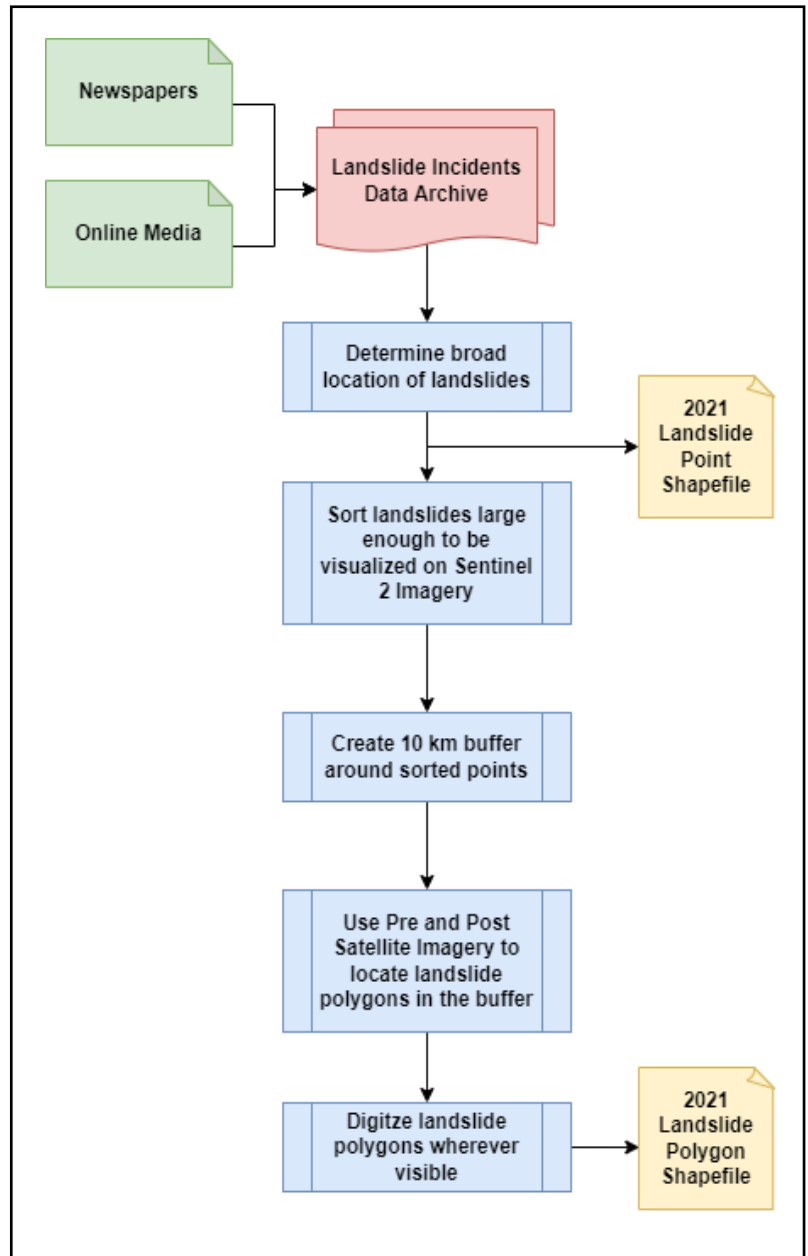


Figure 4: Flowchart for landslide polygon extraction

4.2 AUTOMATIC RAINFALL DATA EXTRACTION

From the list of landslides events, latitude, longitude and date of event are used to extract the corresponding rainfall data using our created automation pipeline. This automation pipeline will first download the required rainfall data from specific sources (Currently GPM data supported) and pre-process them to get hourly precipitation and cumulative precipitation of each event and finally generate the rainfall intensity graph for each event.

The flowchart for the methodology behind Automatic Rainfall Data Extraction is shown in Figure 5.

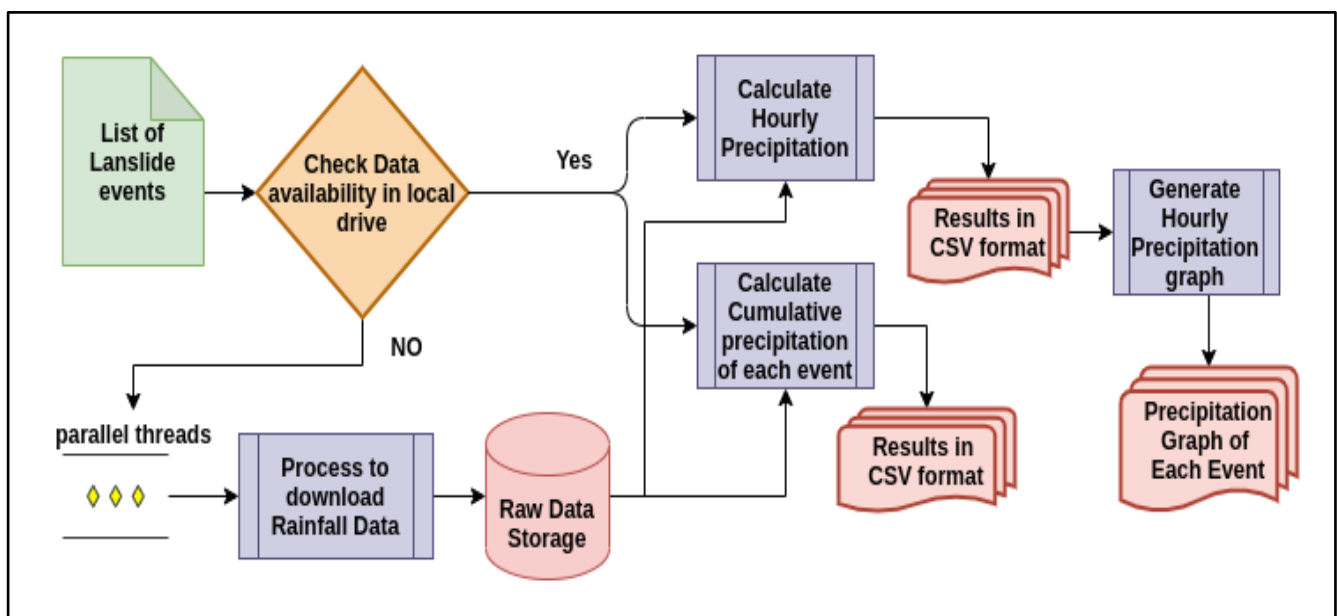


Figure 5: Flowchart for automatic extraction of rainfall

3. LANDSLIDE INVENTORY IN DIFFERENT PARTS OF THE NER

In this section of the report, a very brief summary of landslide incidents in the different states of the NER is provided. Detailed information regarding each landslide incident is provided in the annexure.

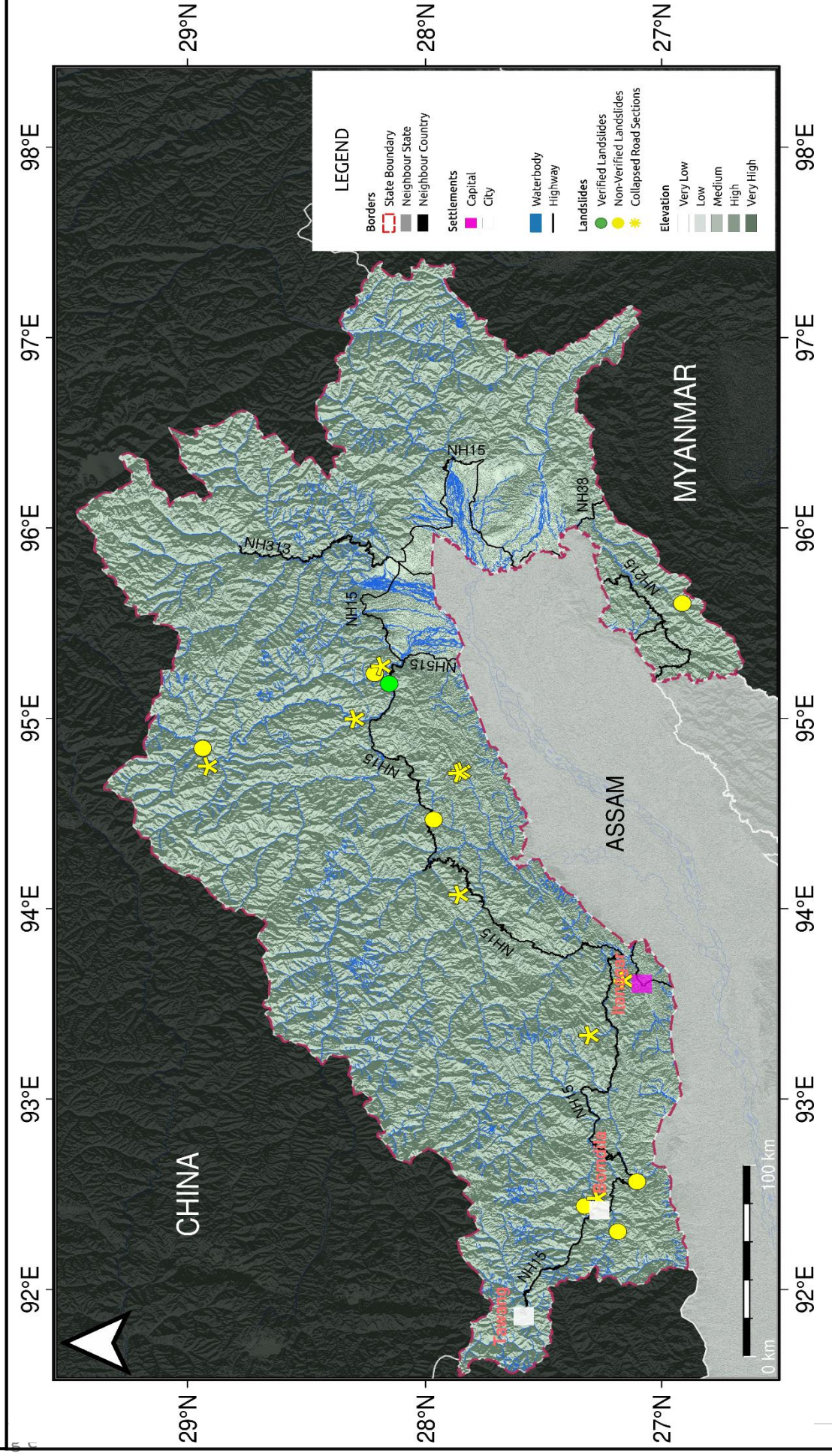
3.1 ARUNACHAL PRADESH

- A total of 17 landslide incidents recorded in Arunachal Pradesh.
- The first landslide incident for the year was recorded on 28th April 2021 and the last event was recorded on 20th September 2021.
- A total of 3 casualties occurred in Arunachal Pradesh in 2021
- NH-13, NH-52, NH-415 and BCT road were some of the highways/road sections that were affected in the state due to landslides.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Online Media/2021/Arunachal Pradesh/1	Online Media	Arunachal Pradesh	Papum Pare	2021-06-08	3	-
2	Newspapers/2021/Arunachal Pradesh/2	Newspapers	Arunachal Pradesh	Upper Siang	2021-08-13	2	-
3	Newspapers/2021/Arunachal Pradesh/3	Newspapers	Arunachal Pradesh	East Siang	2021-08-13	3	-
4	Newspapers/2021/Arunachal Pradesh/4	Newspapers	Arunachal Pradesh	West Siang	2021-08-13	3	-
5	Online Media/2021/Arunachal Pradesh/5	Online Media	Arunachal Pradesh	East Siang	2021-06-27	1	-
6	Online Media/2021/Arunachal Pradesh/6	Online Media	Arunachal Pradesh	Upper Siang	2021-06-17	2	-
7	Newspaper/2021/Arunachal Pradesh/7	Newspaper	Arunachal Pradesh	Upper Subansiri	2021-07-08	2	-
8	Newspaper/2021/Arunachal Pradesh/8	Newspaper	Arunachal Pradesh	West Kameng	2021-06-05	2	2
9	Online Media/2021/Arunachal Pradesh/9	Online Media	Arunachal Pradesh	Papum Pare	2021-05-31	2	-
10	Newspaper/2021/Arunachal Pradesh/10	Newspaper	Arunachal Pradesh	West Kameng	2021-05-25	2	-
11	Newspaper/2021/Arunachal Pradesh/11	Newspaper	Arunachal Pradesh	Tirap	2021-06-22	2	1
12	Online Media/2021/Arunachal Pradesh/12	Online Media	Arunachal Pradesh	Upper Siang	2021-06-30	2	-
13	Online Media/2021/Arunachal Pradesh/13	Online Media	Arunachal Pradesh	West Siang	2021-06-30	3	-
14	Newspaper/2021/Arunachal Pradesh/14	Newspaper	Arunachal Pradesh	West Kameng	2021-04-28	2	-
15	Online Media/2021/Arunachal Pradesh/15	Online Media	Arunachal Pradesh	East Siang	2021-06-27	2	-
16	Newspaper/2021/Arunachal Pradesh/16	Newspaper	Arunachal Pradesh	Lower Subansiri	2021-09-09	2	-
17	Newspaper/2021/Arunachal Pradesh/17	Newspaper	Arunachal Pradesh	West Kameng	2021-09-20	2	-

Table 3: Summary of landslide incidents in Arunachal Pradesh

Landslide Locations 2021, Arunachal Pradesh



Map 2: Landslide Incidents in Arunachal Pradesh, 2021

3.2 ASSAM

- A total of 4 landslide incidents recorded in Assam.
- The first landslide incident for the year was recorded on 18th June 2021 and the last event was recorded on 20th October 2021.
- A total of 2 casualties occurred in Assam in 2021
- NH 39 was closed on one side after the Burapahar Landslide on 25th August 2021.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Online Media/2021/Assam/72	Online Media	Assam	Kamrup metro	2021-06-18	2	1
2	Newspaper/2021/Assam/73	Newspaper	Assam	Nagaon	2021-08-25	3	-
3	Newspaper/2021/Assam/74	Newspaper	Assam	Kamrup metro	2021-10-20	2	-
4	Newspaper/2021/Assam/75	Newspaper	Assam	Kamrup metro	2021-08-06	2	1

Table 4: Summary of landslide incidents in Assam

Landslide Locations 2021, Assam



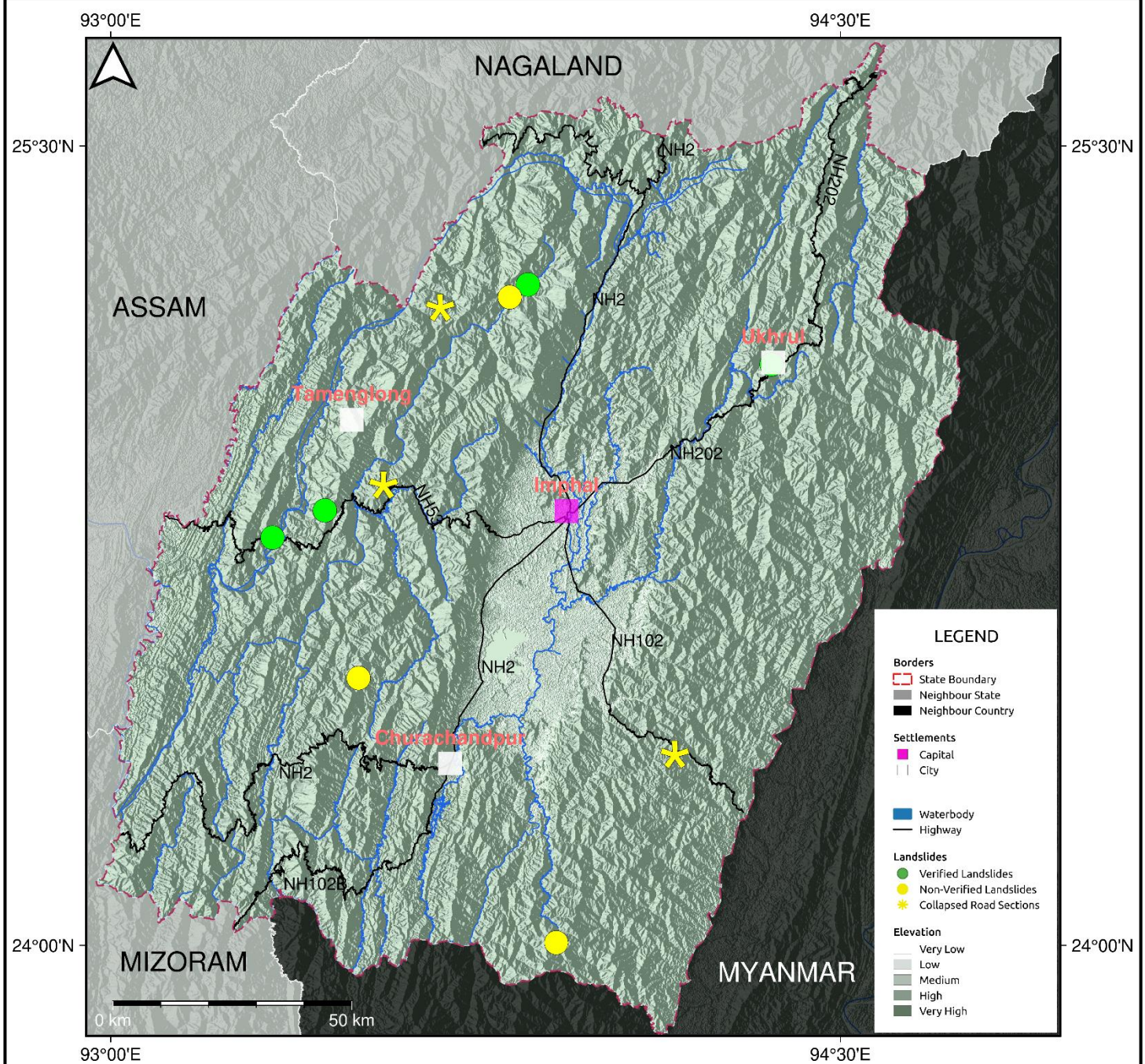
3.3 MANIPUR

- A total of 11 landslide incidents recorded in Manipur.
- The first landslide incident for the year was recorded on 6th June 2021 and the last event was recorded on 9th October 2021.
- A total of 3 casualties occurred in Manipur in 2021
- NH-37, AH-1 and the IT (Imphal-Tamenglong Road) were affected due to landslides this year in the state.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Online Media/2021/Manipur/22	Online Media	Manipur	Senapati	2021-06-06	2	-
2	Newspaper/2021/Manipur/23	Newspaper	Manipur	Tamenglong	2021-08-02	1	-
3	Newspaper/2021/Manipur/24	Newspaper	Manipur	Churachandpur	2021-08-20	2	-
4	Newspaper/2021/Manipur/25	Newspaper	Manipur	Tamenglong	2021-08-16	3	-
5	Online Media/2021/Manipur/26	Online Media	Manipur	Tamenglong	2021-06-24	3	-
6	Online Media/2021/Manipur/27	Online Media	Manipur	Tamenglong	2021-06-07	3	-
7	Online Media/2021/Manipur/28	Online Media	Manipur	Chandel	2021-07-26	3	-
8	Newspaper/2021/Manipur/29	Newspaper	Manipur	Tamenglong	2021-07-19	1	3
9	Newspaper/2021/Manipur/30	Newspaper	Manipur	Ukhrul	2021-06-16	1	-
10	Online Media/2021/Manipur/31	Online Media	Manipur	Tamenglong	2021-07-19	1	-
11	Newspaper/2021/Manipur/32	Newspaper	Manipur	Chandel	2021-10-09	2	-

Table 5: Summary of landslide incidents in Manipur

Landslide Locations 2021, Manipur



Map 4: Landslide Incidents in Manipur 2021

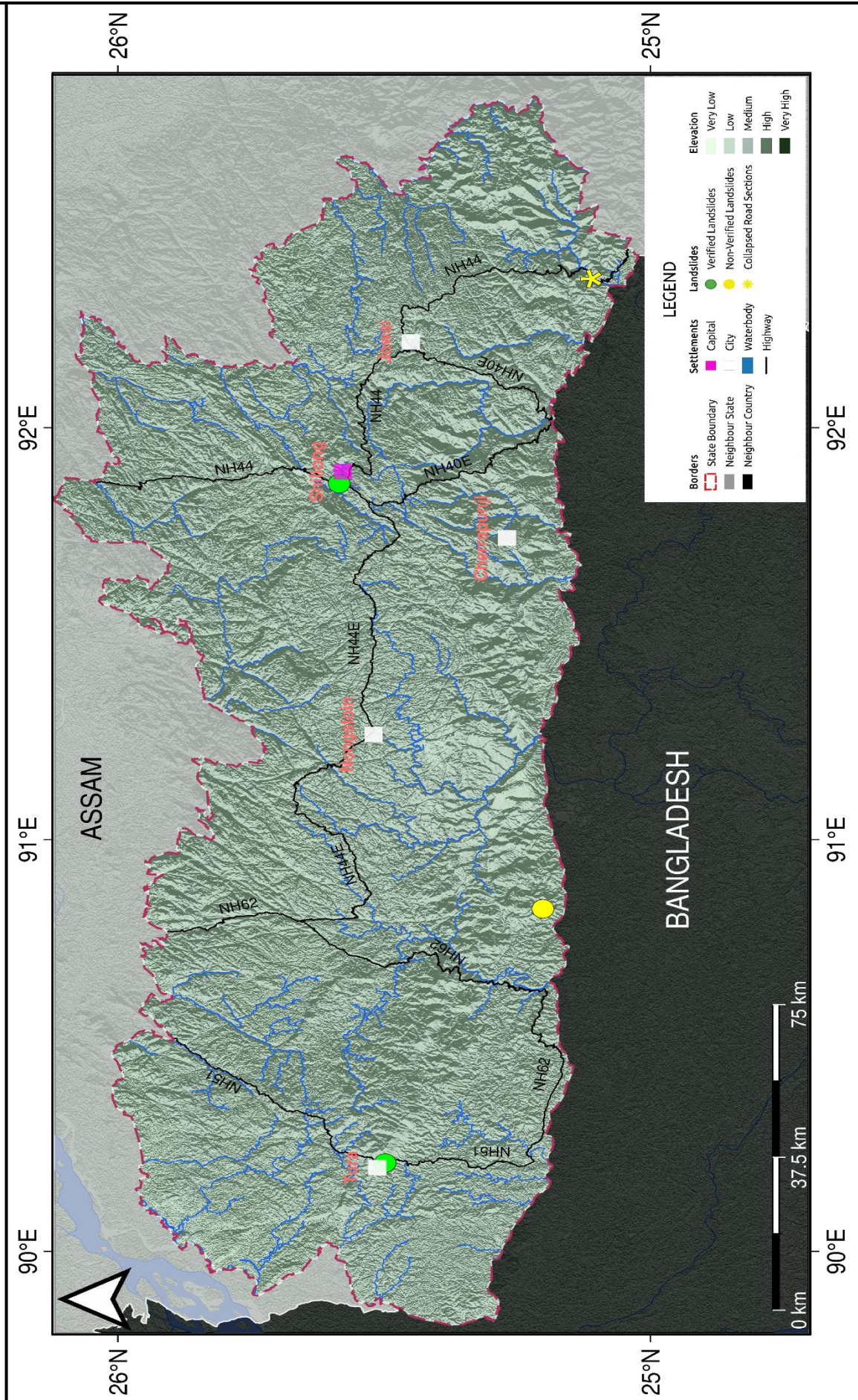
3.4 MEGHALAYA

- A total of 4 landslide incidents recorded Meghalaya.
- The first landslide incident for the year was recorded on 24th February 2021 and the last event was recorded on 2nd August 2021.
- No casualties were reported from Meghalaya in 2021.
- The Sonapani Hydel Project that supplies electricity to the state was affected by landslides this year.
- NH 51 and NH 06 were some of the highways/road sections that were affected in the state due to landslides

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Newspaper/2021/Meghalaya/18	Newspaper	Meghalaya	East Khasi Hills	2021-02-24	1	-
2	Newspaper/2021/Meghalaya/19	Newspaper	Meghalaya	West Garo Hills	2021-06-30	1	-
3	Online Media/2021/Meghalaya/20	Online Media	Meghalaya	Jaintia Hills	2021-08-02	3	-
4	Online Media/2021/Meghalaya/21	Online Media	Meghalaya	South Garo Hills	2021-06-30	2	-

Table 6: Summary of landslide incidents in Meghalaya

Landslide Locations 2021, Meghalaya



Map 5: Landslide Incidents in Meghalaya, 2021

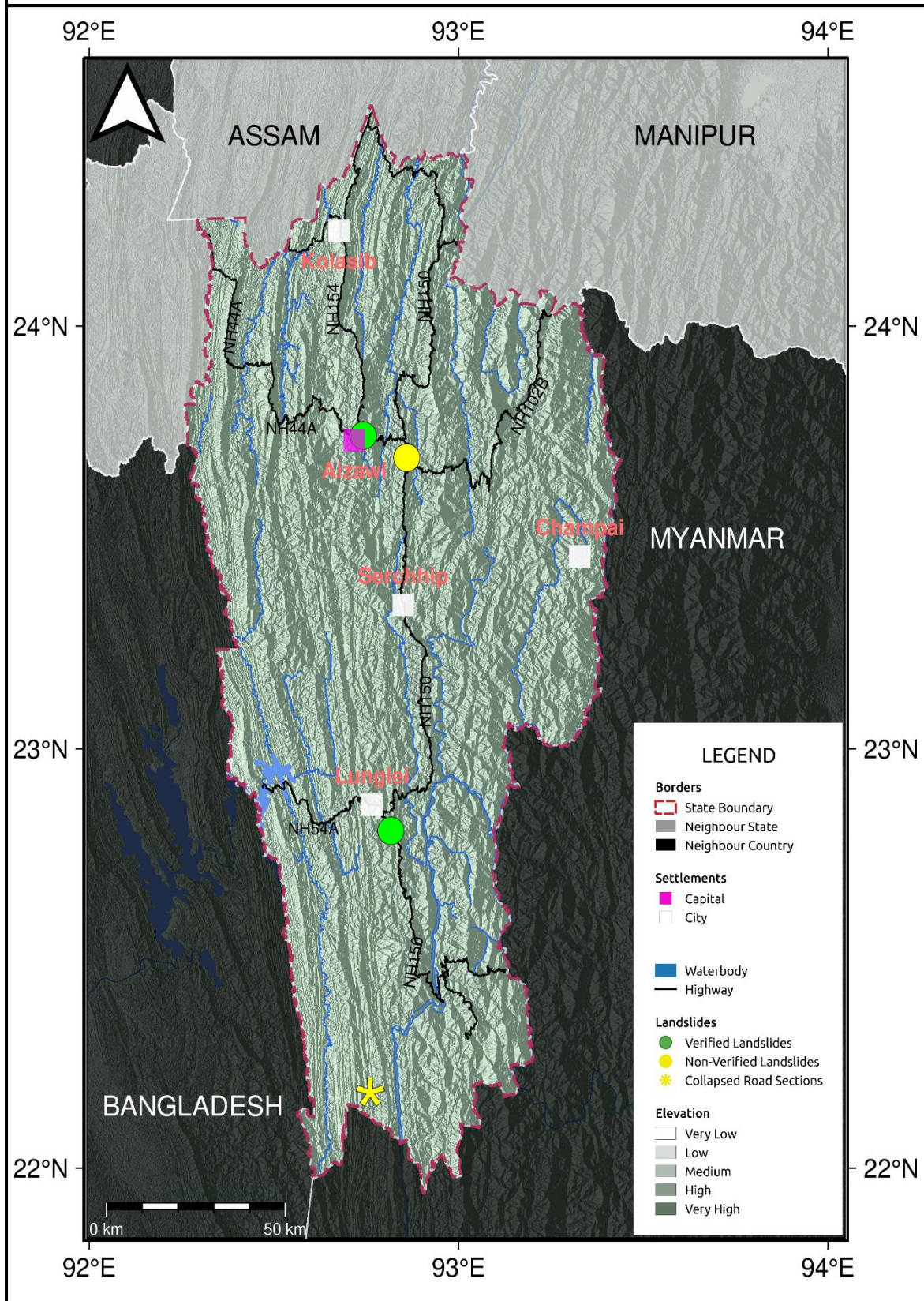
3.5 MIZORAM

- A total of 4 landslide incidents recorded in Mizoram.
- The first landslide incident for the year was recorded on 19th July 2021 and the last event was recorded on 9^h October 2021.
- A total of 6 casualties occurred in Mizoram in 2021.
- Out of the 6 casualties, 4 occurred during a single landslide incident in the Thuampui neighborhood in Aizawl on 11th June 2021.
- NH-54 was affected by landslides in Mizoram in 2021.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Online Media/2021/Mizoram/33	Online Media	Mizoram	Lawngtlai	2021-08-26	3	-
2	Online Media/2021/Mizoram/34	Online Media	Mizoram	Lunglei	2021-07-19	1	-
3	Newspaper/2021/Mizoram/35	Newspaper	Mizoram	Aizawl	2021-06-11	1	4
4	Online Media/2021/Mizoram/36	Online Media	Mizoram	Aizawl	2021-10-09	2	2

Table 7: Summary of landslide incidents in Mizoram

Landslide Locations 2021, Mizoram



Map 6: Landslide Incidents in Mizoram, 2021

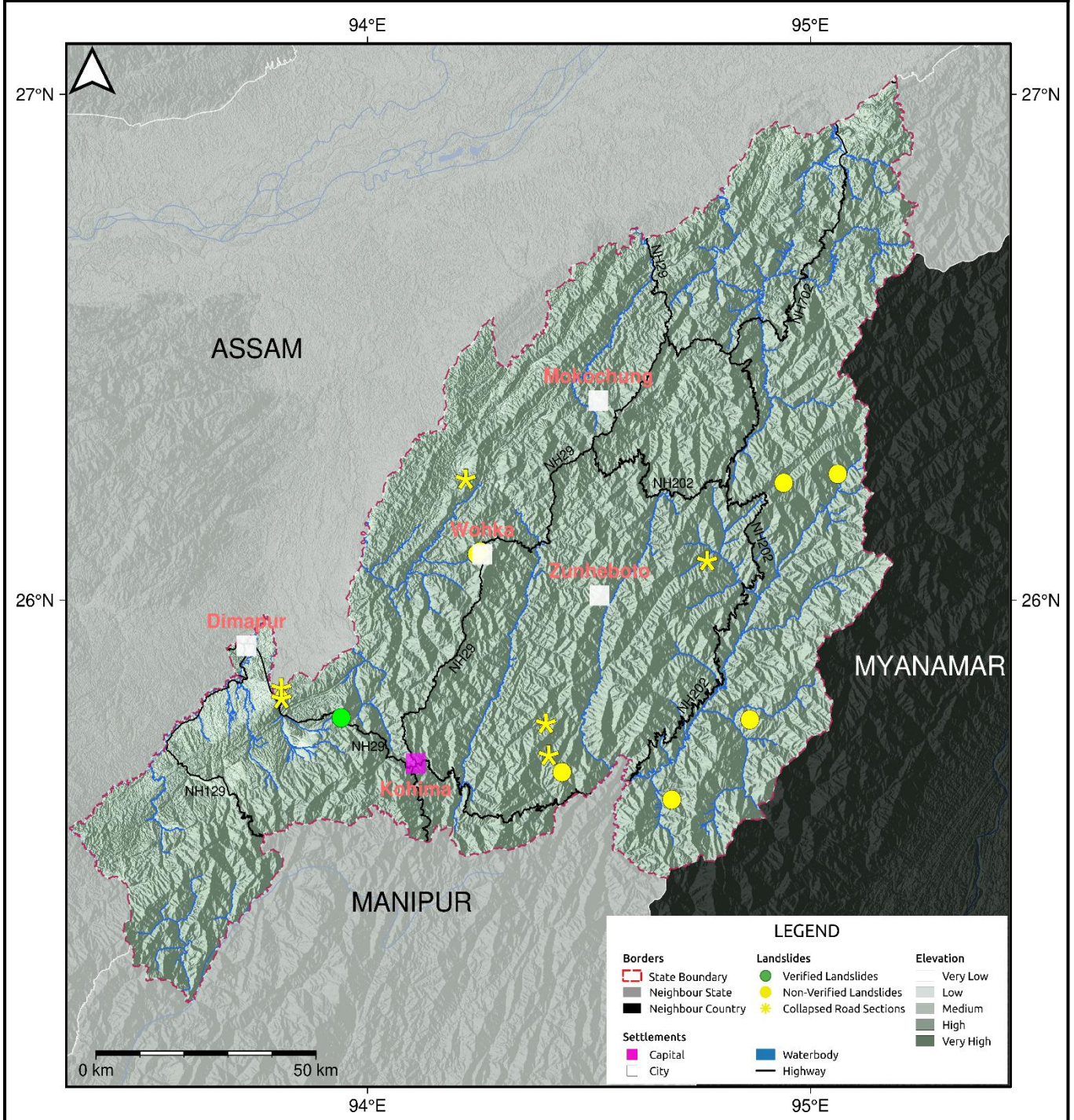
3.6 NAGALAND

- A total of 15 landslide incidents recorded in Nagaland
- The first landslide incident for the year was recorded on 2nd February 2021 and the last event was recorded on 9th September 2021.
- No casualties were reported from Nagaland in 2021
- NH 29 near Chumoukedima Police Gate was affected by landslides and rock-falls due to construction of 4 lane road. Roads near Pfutsero were also affected by landslides in the month of July.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Newspaper/2021/Nagland/37	Newspaper	Nagaland	Dimapur	2021-02-02	3	-
2	Online Media/2021/Nagland/38	Online Media	Nagaland	Dimapur	2021-02-17	3	-
3	Newspaper/2021/Nagland/39	Newspaper	Nagaland	Wokha	2021-08-06	2	-
4	Newspaper/2021/Nagland/40	Newspaper	Nagaland	Tuensang	2021-07-20	3	-
5	Newspaper/2021/Nagland/41	Newspaper	Nagaland	Phek	2021-07-12	2	-
6	Newspaper/2021/Nagland/42	Newspaper	Nagaland	Wokha	2021-08-06	2	-
7	Newspaper/2021/Nagland/43	Newspaper	Nagaland	Wokha	2021-08-06	2	-
8	Newspaper/2021/Nagland/44	Newspaper	Nagaland	Wokha	2021-08-05	3	-
9	Newspaper/2021/Nagland/45	Newspaper	Nagaland	Dimapur	2021-05-10	1	-
10	Online Media/2021/Nagland/46	Online Media	Nagaland	Kiphire	2021-08-17	2	-
11	Online Media/2021/Nagland/47	Online Media	Nagaland	Phek	2021-07-14	2	-
12	Newspaper/2021/Nagland/48	Newspaper	Nagaland	Phek	2021-07-31	3	-
13	Newspaper/2021/Nagland/49	Newspaper	Nagaland	Phek	2021-07-31	3	-
14	Online Media/2021/Nagland/50	Online Media	Nagaland	Tuensang	2021-11-08	2	-
15	Online Media/2021/Nagland/51	Online Media	Nagaland	Tuensang	2021-09-09	2	-

Table 8: Summary of landslide incidents in Nagaland

Landslide Locations 2021, Nagaland



Map 7: Landslide Incidents in Nagaland, 2021

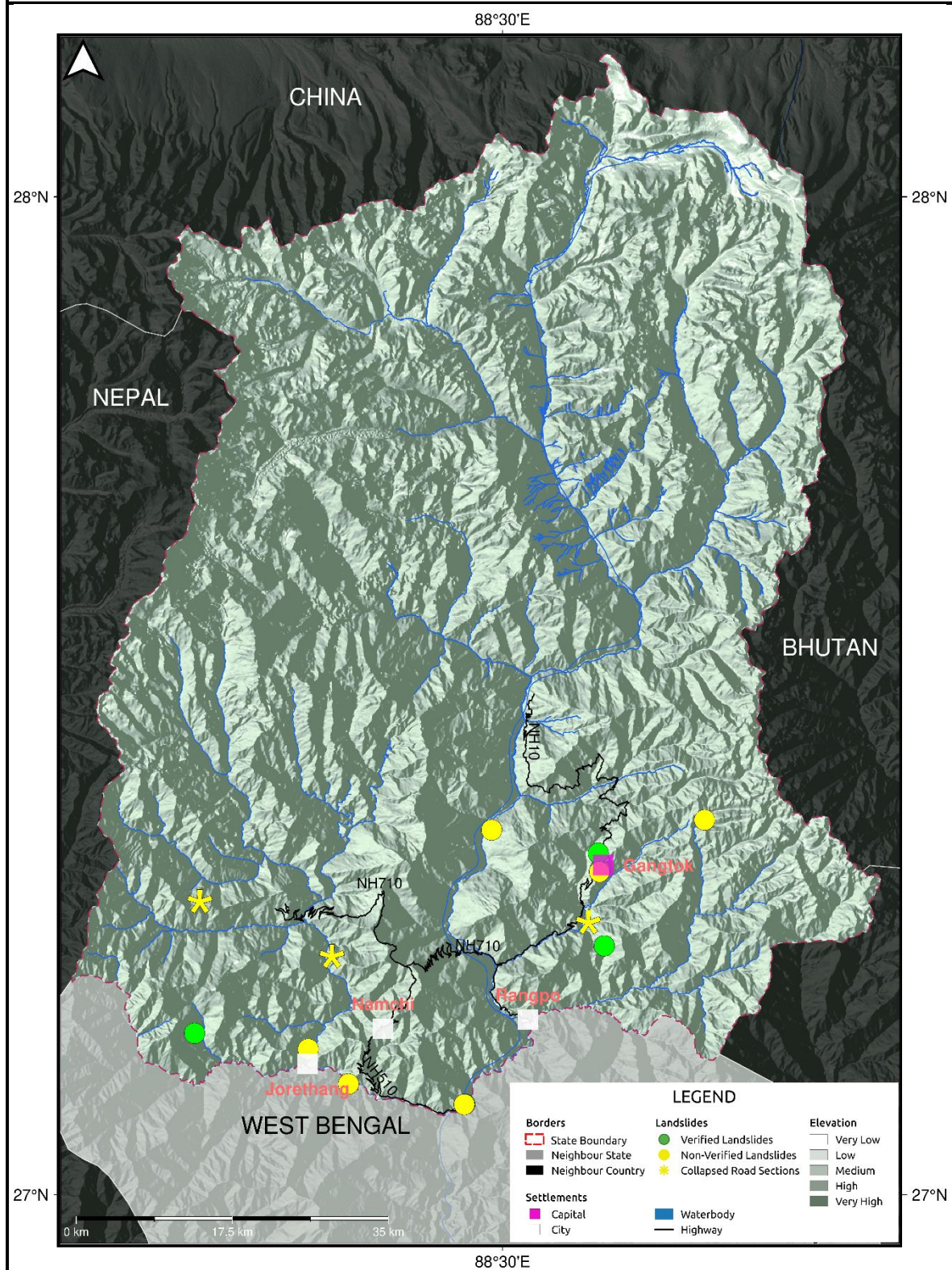
3.7 SIKKIM

- A total of 18 landslide incidents recorded in Sikkim and bordering areas with West Bengal.
- The first landslide incident for the year was recorded on 25th March 2021 and the last event was recorded on 20th October 2021.
- A total of 6 casualties were reported from Sikkim in 2021.
- Melli Bazaar an area on the border of South Sikkim and West Bengal was damaged greatly by landslides on 30th July 2021.
- The roads connecting Pakyong to Assam Lingzey, Jorethang to Legship, Bangten to Chongay and Soreng-Sombaria were affected by landslides.

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Newspaper/2021/Sikkim/52	Newspaper	Sikkim	East Sikkim	2021-08-24	1	-
2	Newspaper/2021/Sikkim/53	Newspaper	Sikkim	West Sikkim	2021-08-24	1	-
3	Newspaper/2021/Sikkim/54	Newspaper	Sikkim	South Sikkim	2021-07-29	1	-
4	Newspaper/2021/Sikkim/55	Newspaper	Sikkim	South Sikkim	2021-07-30	2	1
5	Newspaper/2021/Sikkim-West Bengal/56	Newspaper	Sikkim/West Bengal	Kalimpong	2021-07-30	2	2 (WB)
6	Newspaper/2021/Sikkim-West Bengal/57	Newspaper	Sikkim/West Bengal	Kalimpong	2021-07-30	2	-
7	Newspaper/2021/Sikkim-West Bengal/58	Newspaper	Sikkim/West Bengal	South Sikkim	2021-07-30	2	-
8	Online Media/2021/Sikkim/59	Online Media	Sikkim	East Sikkim	2021-10-20	2	-
9	Newspaper/2021/Sikkim-West Bengal/60	Newspaper	Sikkim/West Bengal	Kalimpong	2021-06-18	2	2 (WB)
10	Online Media/2021/Sikkim/61	Online Media	Sikkim	East Sikkim	2021-03-25	2	-
11	Online Media/2021/Sikkim/62	Online Media	Sikkim	East Sikkim	2021-08-08	3	-
12	Online Media/2021/Sikkim/63	Online Media	Sikkim	East Sikkim	2021-07-31	2	-
13	Online Media/2021/Sikkim-West Bengal/64	Online Media	Sikkim/West Bengal	Kalimpong	2021-07-31	2	-
14	Online Media/2021/Sikkim/65	Online Media	Sikkim	West Sikkim	2021-07-30	2	-
15	Online Media/2021/Sikkim/66	Online Media	Sikkim	West Sikkim	2021-08-27	3	-
16	Newspaper/2021/Sikkim/67	Newspaper	Sikkim	East Sikkim	2021-06-01	1	-
17	Newspaper/2021/Sikkim/68	Newspaper	Sikkim	East Sikkim	2021-06-09	1	1
18	Newspaper/2021/Sikkim/69	Newspaper	Sikkim	West Sikkim	2021-08-29	3	-

Table 9: Summary of landslide incidents in Sikkim

Landslide Locations 2021, Sikkim



Map 8: Landslide Incidents in Sikkim, 2021

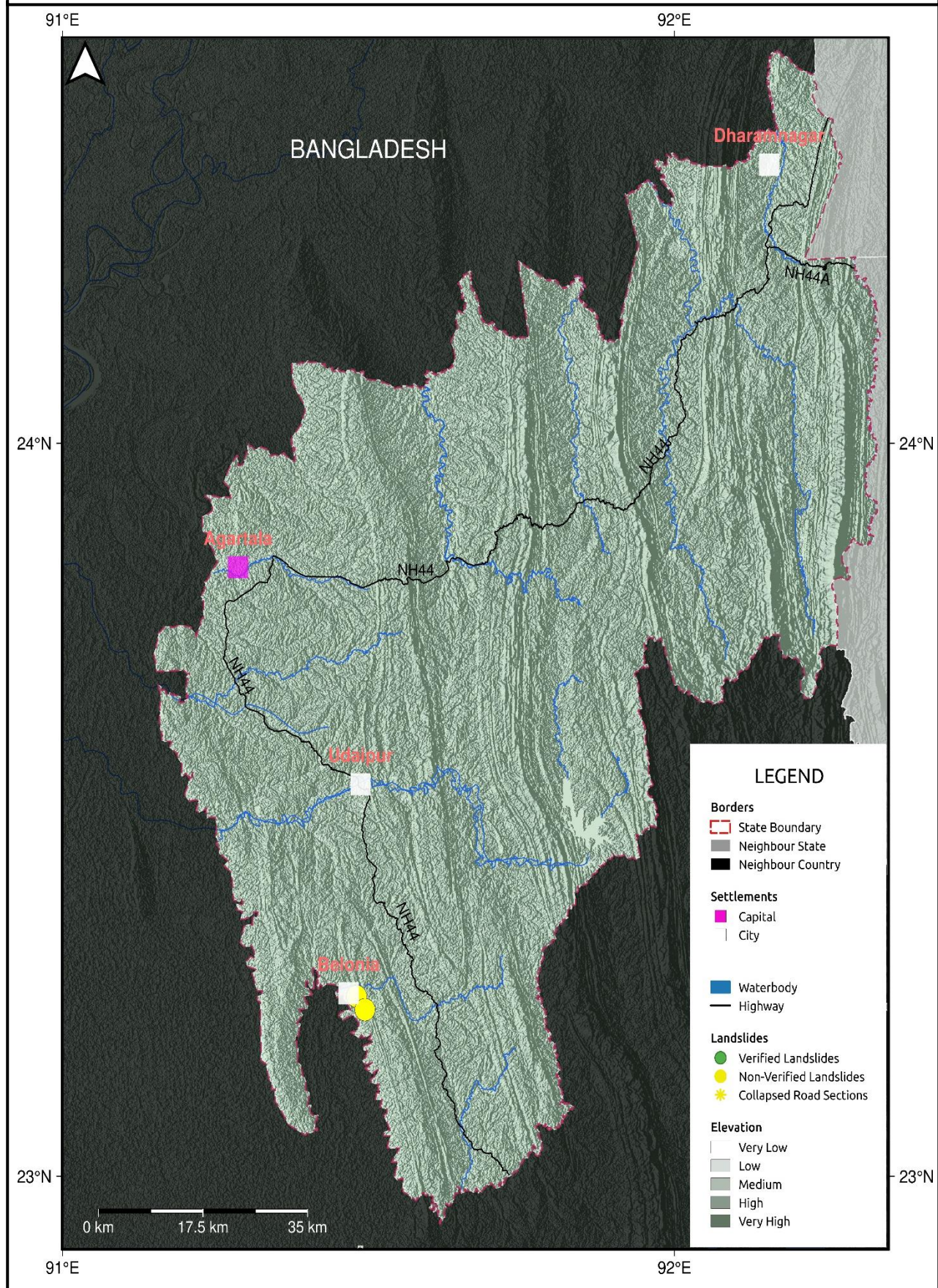
3.8 TRIPURA

- A total of 2 landslide incidents recorded in Tripura.
- Both incidents occurred on 26th August 2021.
- 1 casualty occurred in Tripura in 2021

Sr No	Landslide ID	Source	State	District	Date	Type	Casualty
1	Newspaper/2021/Tripura/70	Newspaper	Tripura	South	2021-08-26	2	1
2	Online Media/2021/Tripura/71	Online Media	Tripura	South	2021-08-26	2	-

Table 10: Summary of landslide incidents in Tripura

Landslide Locations 2021, Tripura



Map 9: Landslide Incidents in Tripura, 2021

ANNEXURES

ANNEXURE-I: LANDSLIDE INCIDENT REPORTS

ARUNACHAL PRADESH

Slide ID	Online Media/2021/Arunachal Pradesh/1
Source	Online Media
Slide No	1
State	Arunachal Pradesh
District	Papum Pare
Name	Leporiang Landslide
Location	Leporiang
NH/SH affected	NH 13
Lat(dd)	27.241524
Lon(dd)	93.335972
Date of Event	2021-06-08
Time of Event	morning
Type of Event	3
Casualty	-
Rainfall(mm)	150.769570

Slide ID	Newspapers/2021/Arunachal Pradesh/2
Source	Newspapers
Slide No	2
State	Arunachal Pradesh
District	Upper Siang
Name	Tuting_Yingkiong_landslide
Location	Between Tuting and Yingkiong
NH/SH affected	Yingkiong Road
Lat(dd)	28.849609
Lon(dd)	94.748649
Date of Event	2021-08-13
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	75.210383

Slide ID	Newspapers/2021/Arunachal Pradesh/3
Source	Newspapers
Slide No	3
State	Arunachal Pradesh
District	East Siang
Name	Sirki Waterfall Landslide
Location	near Sirki waterfall , 4kms from Pasighat towards Pangin and Aalo
NH/SH affected	NH 13
Lat(dd)	28.122053
Lon(dd)	95.274842
Date of Event	2021-08-13
Time of Event	14:48
Type of Event	3
Casualty	-
Rainfall(mm)	52.842427

Slide ID	Newspapers/2021/Arunachal Pradesh/4
Source	Newspapers
Slide No	4
State	Arunachal Pradesh
District	West Siang
Name	Likabali_Basar_landslide
Location	Between Likabali and Basar
NH/SH affected	NH 13
Lat(dd)	27.784222
Lon(dd)	94.721322
Date of Event	2021-08-13
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	19.044726

Slide ID	Online Media/2021/Arunachal Pradesh/5
Source	Online Media
Slide No	5
State	Arunachal Pradesh
District	East Siang
Name	Rotlung Landslide
Location	Rotlung Village at Kebang
NH/SH affected	-
Lat(dd)	28.153611
Lon(dd)	95.181389
Date of Event	2021-06-27
Time of Event	-
Type of Event	1
Casualty	-
Rainfall(mm)	47.738895

Slide ID	Online Media/2021/Arunachal Pradesh/6
Source	Online Media
Slide No	6
State	Arunachal Pradesh
District	Upper Siang
Name	Pekong Landslide
Location	Pekong Village
NH/SH affected	-
Lat(dd)	28.936253
Lon(dd)	94.842537
Date of Event	2021-06-17
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	2.636578

Slide ID	Newspaper/2021/Arunachal Pradesh/7
Source	Newspaper
Slide No	7
State	Arunachal Pradesh
District	Upper Subansiri
Name	Tode landslide
Location	Tode Village
NH/SH affected	Trans Arunachal Highway
Lat(dd)	27.965813
Lon(dd)	94.467809
Date of Event	2021-07-08
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	73.121865

Slide ID	Newspaper/2021/Arunachal Pradesh/8
Source	Newspaper
Slide No	8
State	Arunachal Pradesh
District	West Kameng
Name	BCT Road Landslide
Location	77.200 km in Bhalukpong-Charduar-Tawang (BCT) road near Saddle, Nechipu
NH/SH affected	BCT Road
Lat(dd)	27.326763
Lon(dd)	92.438010
Date of Event	2021-06-05
Time of Event	9:30
Type of Event	2
Casualty	2
Rainfall(mm)	0.000000

Slide ID	Online Media/2021/Arunachal Pradesh/9
Source	Online Media
Slide No	9
State	Arunachal Pradesh
District	Papum Pare
Name	IG Park Landslide
Location	Near Indra Gandhi Park entry gate, Itanagar
NH/SH affected	NH 415
Lat(dd)	27.095983
Lon(dd)	93.622322
Date of Event	2021-05-31
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	69.527662

Slide ID	Newspaper/2021/Arunachal Pradesh/10
Source	Newspaper
Slide No	10
State	Arunachal Pradesh
District	West Kameng
Name	Nechipu_Saddle_landslide
Location	80.3-km between Nechipu and Saddle
NH/SH affected	BCT Road
Lat(dd)	27.208119
Lon(dd)	92.479767
Date of Event	2021-05-25
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	0.010557

Slide ID	Newspaper/2021/Arunachal Pradesh/11
Source	Newspaper
Slide No	11
State	Arunachal Pradesh
District	Tirap
Name	Kuthong Landslide
Location	Kuthong Village in Tirap District
NH/SH affected	-
Lat(dd)	26.911944
Lon(dd)	95.603611
Date of Event	2021-06-22
Time of Event	1:30
Type of Event	2
Casualty	1
Rainfall(mm)	96.874331

Slide ID	Online Media/2021/Arunachal Pradesh/12
Source	Online Media
Slide No	12
State	Arunachal Pradesh
District	Upper Siang
Name	Pasighat_Yingkiong_Landslide
Location	Pasighat to Yignkiong (100 kms from Yingkiong)
NH/SH affected	NH 52
Lat(dd)	28.214774
Lon(dd)	95.235101
Date of Event	2021-06-30
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	29.255691

Slide ID	Online Media/2021/Arunachal Pradesh/13
Source	Online Media
Slide No	13
State	Arunachal Pradesh
District	West Siang
Name	Garu Landslide
Location	near Garu, 25 km from Likabali
NH/SH affected	NH 52
Lat(dd)	27.796145
Lon(dd)	94.713350
Date of Event	2021-06-30
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	18.777780

Slide ID	Newspaper/2021/Arunachal Pradesh/14
Source	Newspaper
Slide No	14
State	Arunachal Pradesh
District	West Kameng
Name	Membachur_Garbow_landslide
Location	between Membachur and Garbow villages
NH/SH affected	-
Lat(dd)	27.186597
Lon(dd)	92.303984
Date of Event	2021-04-28
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	0.538725

Slide ID	Online Media/2021/Arunachal Pradesh/15
Source	Online Media
Slide No	15
State	Arunachal Pradesh
District	East Siang
Name	Sangam_bridge_collapse
Location	Sangam Bridge Collapse near Pangin
NH/SH affected	-
Lat(dd)	28.233722
Lon(dd)	94.996300
Date of Event	2021-06-27
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	47.070611

Slide ID	Newspaper/2021/Arunachal Pradesh/16
Source	Newspaper
Slide No	16
State	Arunachal Pradesh
District	Lower Subansiri
Name	Raga landslide
Location	Raga headquarters
NH/SH affected	-
Lat(dd)	27.794025
Lon(dd)	94.073954
Date of Event	2021-09-09
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	3.679493

Slide ID	Newspaper/2021/Arunachal Pradesh/17
Source	Newspaper
Slide No	17
State	Arunachal Pradesh
District	West Kameng
Name	20_landslides_BCT_road
Location	20 locations along BCT Road between Sessa and Nechippu
NH/SH affected	-
Lat(dd)	27.104442
Lon(dd)	92.566072
Date of Event	2021-09-20
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	0.000000

ASSAM

Slide ID	Online Media/2021/Assam/72
Source	Online Media
Slide No	72
State	Assam
District	Kamrup metro
Name	Water_works_colony_landslide
Location	Water Works Colony
NH/SH affected	-
Lat(dd)	26.162784
Lon(dd)	91.694636
Date of Event	2021-06-18
Time of Event	4:30
Type of Event	2
Casualty	1
Rainfall(mm)	121.189373

Slide ID	Newspaper/2021/Assam/73
Source	Newspaper
Slide No	73
State	Assam
District	Nagaon
Name	Burapahar Landslide
Location	Kaliabor
NH/SH affected	NH 39
Lat(dd)	26.574761
Lon(dd)	93.110996
Date of Event	2021-08-25
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	37.999535

Slide ID	Newspaper/2021/Assam/74
Source	Newspaper
Slide No	74
State	Assam
District	Kamrup metro
Name	Pandu_landslide
Location	Guwahati
NH/SH affected	-
Lat(dd)	26.170251
Lon(dd)	91.690161
Date of Event	2021-10-20
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	21.309880

Slide ID	Newspaper/2021/Assam/75
Source	Newspaper
Slide No	75
State	Assam
District	Kamrup metro
Name	Borabari_landslide
Location	Borbari
NH/SH affected	-
Lat(dd)	26.154238
Lon(dd)	91.810523
Date of Event	2021-08-06
Time of Event	14:30
Type of Event	2
Casualty	1
Rainfall(mm)	0.009634

MANIPUR

Slide ID	Online Media/2021/Manipur/22
Source	Online Media
Slide No	22
State	Manipur
District	Senapati
Name	Tujang_Waichong_landslide
Location	Between Tujang Waichong and Selsi Village
NH/SH affected	-
Lat(dd)	25.217562
Lon(dd)	93.820138
Date of Event	2021-06-06
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	10.335549

Slide ID	Newspaper/2021/Manipur/23
Source	Newspaper
Slide No	23
State	Manipur
District	Tamenglong
Name	Lamlenkon_landslide
Location	Lamlenkon in Tujang Vaichong
NH/SH affected	Imphal-Tamenglong Road
Lat(dd)	25.241119
Lon(dd)	93.857702
Date of Event	2021-08-02
Time of Event	3:30
Type of Event	1
Casualty	-
Rainfall(mm)	7.934080

Slide ID	Newspaper/2021/Manipur/24
Source	Newspaper
Slide No	24
State	Manipur
District	Churachandpur
Name	S_kholep_landslide
Location	S Kholep village
NH/SH affected	Henglep to Churachandpur road
Lat(dd)	24.503658
Lon(dd)	93.509375
Date of Event	2021-08-20
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	101.288447

Slide ID	Newspaper/2021/Manipur/25
Source	Newspaper
Slide No	25
State	Manipur
District	Tamenglong
Name	Awang_khul_landslide
Location	5 km away from Awang Khul Village
NH/SH affected	NH 37
Lat(dd)	24.826041
Lon(dd)	93.561190
Date of Event	2021-08-16
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	4.368737

Slide ID	Online Media/2021/Manipur/26
Source	Online Media
Slide No	26
State	Manipur
District	Tamenglong
Name	Tamei_landslide
Location	Tamei Village
NH/SH affected	Imphal-Tamenglong Road
Lat(dd)	25.158298
Lon(dd)	93.677995
Date of Event	2021-06-24
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	15.714632

Slide ID	Online Media/2021/Manipur/27
Source	Online Media
Slide No	27
State	Manipur
District	Tamenglong
Name	Tamei2_landslide
Location	Tamei Village
NH/SH affected	Imphal-Tamenglong Road
Lat(dd)	25.158298
Lon(dd)	93.677995
Date of Event	2021-06-07
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	20.428739

Slide ID	Online Media/2021/Manipur/28
Source	Online Media
Slide No	28
State	Manipur
District	Chandel
Name	Chahmol_landslide
Location	Chahmol Village
NH/SH affected	AH-1 Moreh Road
Lat(dd)	24.319566
Lon(dd)	94.160246
Date of Event	2021-07-26
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	1.530845

Slide ID	Newspaper/2021/Manipur/29
Source	Newspaper
Slide No	29
State	Manipur
District	Tamenglong
Name	Thingou_landslide
Location	8km towards Thingou, 15 km north from Nungba PS
NH/SH affected	-
Lat(dd)	24.818353
Lon(dd)	93.440246
Date of Event	2021-07-19
Time of Event	14:50
Type of Event	1
Casualty	3
Rainfall(mm)	48.191199

Slide ID	Newspaper/2021/Manipur/30
Source	Newspaper
Slide No	30
State	Manipur
District	Ukhrul
Name	Mini_Secretariat_landslide
Location	Mini Secretariat Complex, Hamleikhong
NH/SH affected	-
Lat(dd)	25.091769
Lon(dd)	94.358421
Date of Event	2021-06-16
Time of Event	-
Type of Event	1
Casualty	-
Rainfall(mm)	0.000000

Slide ID	Online Media/2021/Manipur/31
Source	Online Media
Slide No	31
State	Manipur
District	Tamenglong
Name	Sibilong_temple_landslide
Location	Sibilong Temple
NH/SH affected	Imphal-Jiribam Road (NH-37)
Lat(dd)	24.767366
Lon(dd)	93.332535
Date of Event	2021-07-19
Time of Event	-
Type of Event	1
Casualty	-
Rainfall(mm)	67.985381

Slide ID	Newspaper/2021/Manipur/32
Source	Newspaper
Slide No	32
State	Manipur
District	Chandel
Name	Khulmulen-Sajik landslide
Location	Khulmulen-Sajik Road in Sajik Valley
NH/SH affected	-
Lat(dd)	24.005219
Lon(dd)	93.915829
Date of Event	2021-10-09
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	0.093849

MEGHALAYA

Slide ID	Newspaper/2021/Meghalaya/18
Source	Newspaper
Slide No	18
State	Meghalaya
District	East Khasi Hills
Name	Sonapani_landslide
Location	Sonapani Hydro-Electric Power Project
NH/SH affected	-
Lat(dd)	25.585545
Lon(dd)	91.864660
Date of Event	2021-02-24
Time of Event	-
Type of Event	1
Casualty	-
Rainfall(mm)	0.276384

Slide ID	Newspaper/2021/Meghalaya/19
Source	Newspaper
Slide No	19
State	Meghalaya
District	West Garo Hills
Name	Tura_dalu_stretch
Location	Tura-Dalu stretch
NH/SH affected	NH 51
Lat(dd)	25.499358
Lon(dd)	90.214668
Date of Event	2021-06-30
Time of Event	-
Type of Event	1
Casualty	-
Rainfall(mm)	107.621934

Slide ID	Online Media/2021/Meghalaya/20
Source	Online Media
Slide No	20
State	Meghalaya
District	Jaintia Hills
Name	Kuliang_landslide
Location	Kuliang Village, Khilerihat
NH/SH affected	NH 06
Lat(dd)	25.081742
Lon(dd)	92.362063
Date of Event	2021-08-02
Time of Event	7:00
Type of Event	3
Casualty	-
Rainfall(mm)	90.034167

Slide ID	Online Media/2021/Meghalaya/21
Source	Online Media
Slide No	21
State	Meghalaya
District	South Garo Hills
Name	Malikona_Village_landslide
Location	Malikona Village Baghmara town
NH/SH affected	-
Lat(dd)	25.202693
Lon(dd)	90.831915
Date of Event	2021-06-30
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	200.134458

MIZORAM

Slide ID	Online Media/2021/Mizoram/33
Source	Online Media
Slide No	33
State	Mizoram
District	Lawngtlai
Name	Zochachhuah landslide
Location	Zochachhuah
NH/SH affected	-
Lat(dd)	22.130154
Lon(dd)	92.761693
Date of Event	2021-08-26
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	14.347636

Slide ID	Online Media/2021/Mizoram/34
Source	Online Media
Slide No	34
State	Mizoram
District	Lunglei
Name	Bualtei landslide
Location	Bualtei
NH/SH affected	NH 54
Lat(dd)	22.805047
Lon(dd)	92.816565
Date of Event	2021-07-19
Time of Event	-
Type of Event	1
Casualty	-
Rainfall(mm)	203.521302

Slide ID	Newspaper/2021/Mizoram/35
Source	Newspaper
Slide No	35
State	Mizoram
District	Aizawl
Name	Thuampui landslide
Location	Thuampui
NH/SH affected	-
Lat(dd)	23.741920
Lon(dd)	92.741738
Date of Event	2021-06-11
Time of Event	21:30:00
Type of Event	1
Casualty	4
Rainfall(mm)	17.590322

Slide ID	Online Media/2021/Mizoram/36
Source	Online Media
Slide No	36
State	Mizoram
District	Aizawl
Name	Thingsulthiah landslide
Location	Thingsulthiah Village
NH/SH affected	-
Lat(dd)	23.689863
Lon(dd)	92.859428
Date of Event	2021-10-09
Time of Event	-
Type of Event	2
Casualty	2
Rainfall(mm)	7.244019

NAGALAND

Slide ID	Newspaper/2021/Nagland/37
Source	Newspaper
Slide No	37
State	Nagaland
District	Dimapur
Name	TL Park landslide
Location	Near TL Park between Chumoukedima and Kukidalong
NH/SH affected	-
Lat(dd)	25.793499
Lon(dd)	93.805655
Date of Event	2021-02-02
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	0.000000

Slide ID	Online Media/2021/Nagaland/38
Source	Online Media
Slide No	38
State	Nagaland
District	Dimapur
Name	TL Park 2 landslide
Location	Near TL Park between Chumoukedima and Kukidalong
NH/SH affected	-
Lat(dd)	25.772599
Lon(dd)	93.806000
Date of Event	2021-02-17
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	0.000000

Slide ID	Newspaper/2021/Nagland/39
Source	Newspaper
Slide No	39
State	Nagaland
District	Wokha
Name	Lower Etsuchuka Colony landslide
Location	Lower Etsuchuka Colony
NH/SH affected	-
Lat(dd)	26.092899
Lon(dd)	94.254627
Date of Event	2021-08-06
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	57.994488

Slide ID	Newspaper/2021/Nagland/40
Source	Newspaper
Slide No	40
State	Nagaland
District	Tuensang
Name	Wangshu_PwD_Road_landslide
Location	Wangshu Village
NH/SH affected	PWD Road connecting Tobu to ITC Pangsha
Lat(dd)	26.046657
Lon(dd)	94.767115
Date of Event	2021-07-20
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	43.889829

Slide ID	Newspaper/2021/Nagland/41
Source	Newspaper
Slide No	41
State	Nagaland
District	Phek
Name	Losami_Phek_landslide
Location	Landslide between Losami and Phek Village
NH/SH affected	-
Lat(dd)	25.657788
Lon(dd)	94.438812
Date of Event	2021-07-12
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	1.334883

Slide ID	Newspaper/2021/Nagland/42
Source	Newspaper
Slide No	42
State	Nagaland
District	Wokha
Name	Police_Point_landslide
Location	Police Point
NH/SH affected	-
Lat(dd)	26.096249
Lon(dd)	94.254289
Date of Event	2021-08-06
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	57.994488

Slide ID	Newspaper/2021/Nagland/43
Source	Newspaper
Slide No	43
State	Nagaland
District	Wokha
Name	Airfield_colony_landslide
Location	Airfield Colony
NH/SH affected	-
Lat(dd)	26.091211
Lon(dd)	94.246956
Date of Event	2021-08-06
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	57.994488

Slide ID	Newspaper/2021/Nagland/44
Source	Newspaper
Slide No	44
State	Nagaland
District	Wokha
Name	Sanis_Doyang_landslide
Location	Between Sanis and Doyang
NH/SH affected	-
Lat(dd)	26.208542
Lon(dd)	94.221938
Date of Event	2021-08-05
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	8.485464

Slide ID	Newspaper/2021/Nagland/45
Source	Newspaper
Slide No	45
State	Nagaland
District	Dimapur
Name	PipHEMA_landslide
Location	PipHEMA
NH/SH affected	-
Lat(dd)	25.766187
Lon(dd)	93.940678
Date of Event	2021-05-10
Time of Event	07:00
Type of Event	1
Casualty	-
Rainfall(mm)	2.610091

Slide ID	Online Media/2021/Nagland/46
Source	Online Media
Slide No	46
State	Nagaland
District	Kiphire
Name	Pungro-Mimi landslide
Location	Pungro-Mimi axis near Longkhimong village
NH/SH affected	-
Lat(dd)	25.762300
Lon(dd)	94.863047
Date of Event	2021-08-17
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	59.658672

Slide ID	Online Media/2021/Nagland/47
Source	Online Media
Slide No	47
State	Nagaland
District	Phek
Name	Phek_landslide
Location	Ziphu
NH/SH affected	-
Lat(dd)	25.603106
Lon(dd)	94.686755
Date of Event	2021-07-14
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	65.401161

Slide ID	Newspaper/2021/Nagland/48
Source	Newspaper
Slide No	48
State	Nagaland
District	Phek
Name	Pfutsiro-Khomi-Phek landslide
Location	-
NH/SH affected	Pfutsiro-Khomi-Phek
Lat(dd)	25.659725
Lon(dd)	94.409427
Date of Event	2021-07-31
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	62.610600

Slide ID	Newspaper/2021/Nagland/49
Source	Newspaper
Slide No	49
State	Nagaland
District	Phek
Name	Pfutsiro-Lanye Landslide
Location	-
NH/SH affected	Pfutsiro-Lanye
Lat(dd)	25.723469
Lon(dd)	94.402948
Date of Event	2021-07-31
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	37.740973

Slide ID	Online Media/2021/Nagland/50
Source	Online Media
Slide No	50
State	Nagaland
District	Tuensang
Name	Wensoi Landslide
Location	Wensoi Village
NH/SH affected	-
Lat(dd)	26.250081
Lon(dd)	95.061502
Date of Event	2021-11-08
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	0.000000

Slide ID	Online Media/2021/Nagland/51
Source	Online Media
Slide No	51
State	Nagaland
District	Tuensang
Name	Chedang Saddle Landslide
Location	Chendang Saddle and Noklak
NH/SH affected	-
Lat(dd)	26.232589
Lon(dd)	94.939423
Date of Event	2021-09-09
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	27.739122

SIKKIM

Slide ID	Newspaper/2021/Sikkim/52
Source	Newspaper
Slide No	52
State	Sikkim
District	East Sikkim
Name	Samsing Shanti Turn Landslide
Location	Samsing Shanti turn
NH/SH affected	Pakyong to Assam Lingsey Road
Lat(dd)	27.250224
Lon(dd)	88.614992
Date of Event	2021-08-24
Time of Event	-
Type of Event	1
Casualty	-
Rainfall(mm)	6.267921

Slide ID	Newspaper/2021/Sikkim/53
Source	Newspaper
Slide No	53
State	Sikkim
District	West Sikkim
Name	Muneew Tareybhair Landslide
Location	Muneew Taraybhair stretch
NH/SH affected	Soreng Sombaria Road
Lat(dd)	27.162400
Lon(dd)	88.152087
Date of Event	2021-08-24
Time of Event	-
Type of Event	1
Casualty	-
Rainfall(mm)	8.109331

Slide ID	Newspaper/2021/Sikkim/54
Source	Newspaper
Slide No	54
State	Sikkim
District	South Sikkim
Name	Melli Secondary School Landslide
Location	Melli Secondary School
NH/SH affected	-
Lat(dd)	27.090965
Lon(dd)	88.457440
Date of Event	2021-07-29
Time of Event	-
Type of Event	1
Casualty	-
Rainfall(mm)	33.110832

Slide ID	Newspaper/2021/Sikkim/55
Source	Newspaper
Slide No	55
State	Sikkim
District	South Sikkim
Name	Melli Bazaar Landslide
Location	Melli Bazaar
NH/SH affected	-
Lat(dd)	27.090724
Lon(dd)	88.456816
Date of Event	2021-07-30
Time of Event	-
Type of Event	2
Casualty	1
Rainfall(mm)	95.820024

Slide ID	Newspaper/2021/Sikkim-West Bengal/56
Source	Newspaper
Slide No	56
State	Sikkim/West Bengal
District	Kalimpong
Name	Mamkhola Landslide
Location	Mamkhola
NH/SH affected	-
Lat(dd)	27.142902
Lon(dd)	88.531935
Date of Event	2021-07-30
Time of Event	-
Type of Event	2
Casualty	2 (WB)
Rainfall(mm)	27.444981

Slide ID	Newspaper/2021/Sikkim-West Bengal/57
Source	Newspaper
Slide No	57
State	Sikkim/West Bengal
District	Kalimpong
Name	Melli 2 Landslide
Location	near Melli
NH/SH affected	-
Lat(dd)	27.108710
Lon(dd)	88.466754
Date of Event	2021-07-30
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	95.820024

Slide ID	Newspaper/2021/Sikkim-West Bengal/58
Source	Newspaper
Slide No	58
State	Sikkim/West Bengal
District	South Sikkim
Name	Melli_Jorethang_landslide
Location	Between Melli and Jorethang
NH/SH affected	-
Lat(dd)	27.111336
Lon(dd)	88.325929
Date of Event	2021-07-30
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	63.980069

Slide ID	Online Media/2021/Sikkim/59
Source	Online Media
Slide No	59
State	Sikkim
District	East Sikkim
Name	Pani House Landslide
Location	Pani House
NH/SH affected	-
Lat(dd)	27.324590
Lon(dd)	88.609439
Date of Event	2021-10-20
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	198.109475

Slide ID	Newspaper/2021/Sikkim-West Bengal/60
Source	Newspaper
Slide No	60
State	Sikkim/West Bengal
District	Kalimpong
Name	Bhalukhola_landslide
Location	Bhalu Khola
NH/SH affected	-
Lat(dd)	27.091170
Lon(dd)	88.464695
Date of Event	2021-06-18
Time of Event	-
Type of Event	2
Casualty	2 (WB)
Rainfall(mm)	5.959684

Slide ID	Online Media/2021/Sikkim/61
Source	Online Media
Slide No	61
State	Sikkim
District	East Sikkim
Name	Kyonsola_waterfall_landslide
Location	Kyonsola waterfall on way to Tsongmo
NH/SH affected	-
Lat(dd)	27.376583
Lon(dd)	88.728341
Date of Event	2021-03-25
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	3.300459

Slide ID	Online Media/2021/Sikkim/62
Source	Online Media
Slide No	62
State	Sikkim
District	East Sikkim
Name	Basilakha_landslide
Location	Basilakha
NH/SH affected	Lalturning-Basilakha Road
Lat(dd)	27.255462
Lon(dd)	88.597286
Date of Event	2021-08-08
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	105.199218

Slide ID	Online Media/2021/Sikkim/63
Source	Online Media
Slide No	63
State	Sikkim
District	East Sikkim
Name	Seley_Ward_landslide
Location	Seley ward under Tumin Lingee Constituency
NH/SH affected	-
Lat(dd)	27.366542
Lon(dd)	88.487805
Date of Event	2021-07-31
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	140.635612

Slide ID	Online Media/2021/Sikkim-West Bengal/64
Source	Online Media
Slide No	64
State	Sikkim/West Bengal
District	Kalimpong
Name	Seti Jhora Landslide
Location	Seti Jhora near Kalimpong
NH/SH affected	NH 10
Lat(dd)	26.939387
Lon(dd)	88.441413
Date of Event	2021-07-31
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	29.429714

Slide ID	Online Media/2021/Sikkim/65
Source	Online Media
Slide No	65
State	Sikkim
District	West Sikkim
Name	Sisney Landslide
Location	Sisney Petrol Pump
NH/SH affected	Joerthang - Legship Road
Lat(dd)	27.146369
Lon(dd)	88.280411
Date of Event	2021-07-30
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	24.942364

Slide ID	Online Media/2021/Sikkim/66
Source	Online Media
Slide No	66
State	Sikkim
District	West Sikkim
Name	Jorethang-Legship Landslide
Location	Reshi Bazaar
NH/SH affected	Joerthang - Legship Road
Lat(dd)	27.221268
Lon(dd)	88.307456
Date of Event	2021-08-27
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	55.277454

Slide ID	Newspaper/2021/Sikkim/67
Source	Newspaper
Slide No	67
State	Sikkim
District	East Sikkim
Name	Seesa Golai Landslide
Location	Seesa Golai
NH/SH affected	-
Lat(dd)	27.326815
Lon(dd)	88.610230
Date of Event	2021-06-01
Time of Event	-
Type of Event	1
Casualty	-
Rainfall(mm)	50.510061

Slide ID	Newspaper/2021/Sikkim/68
Source	Newspaper
Slide No	68
State	Sikkim
District	East Sikkim
Name	Tashi Namgyal Secondary School Landslide
Location	-
NH/SH affected	-
Lat(dd)	27.343272
Lon(dd)	88.608275
Date of Event	2021-06-09
Time of Event	-
Type of Event	1
Casualty	1
Rainfall(mm)	66.071349

Slide ID	Newspaper/2021/Sikkim/69
Source	Newspaper
Slide No	69
State	Sikkim
District	West Sikkim
Name	Bongten-Changay Landslide
Location	Upper Bongten and Changay
NH/SH affected	Bongten-Changay road
Lat(dd)	27.276428
Lon(dd)	88.158235
Date of Event	2021-08-29
Time of Event	-
Type of Event	3
Casualty	-
Rainfall(mm)	12.638664

TRIPURA

Slide ID	Newspaper/2021/Tripura/70
Source	Newspaper
Slide No	70
State	Tripura
District	South
Name	Sarsima Landslide
Location	Sarsima area of Belonia in South Tripura
NH/SH affected	-
Lat(dd)	23.246300
Lon(dd)	91.480725
Date of Event	2021-08-26
Time of Event	-
Type of Event	2
Casualty	1
Rainfall(mm)	78.381047

Slide ID	Online Media/2021/Tripura/71
Source	Online Media
Slide No	71
State	Tripura
District	South
Name	Jharjhari landslide
Location	Jharjhari Village
NH/SH affected	-
Lat(dd)	23.228158
Lon(dd)	91.494288
Date of Event	2021-08-26
Time of Event	-
Type of Event	2
Casualty	-
Rainfall(mm)	78.381047

ANNEXURE-II: NEWSPAPER (CLIPPINGS & DIGITAL ARCHIVE)

Summit Times 03/09/21

SDM visits landslide affected Upper Bongten



RUDRA KAUSHIK GANTOK, 02 SEPT: SDM, Dentam, Nanda Kumar Karki visited Upper Bongten today to take stock of damages incurred in landslide that occurred on 29 Aug that washed away the road connecting Changay with Bongten and other important areas. He assured all possible help from the government and instructed the concerned AE and other officials to take the required action. The road has been made motorable temporarily by the road supervisor and contractor with the help of villagers. People of Upper Bongten and Changay (Upper Sapung) took this opportunity to thank SDM, Dentam and digital media "Hamro Khabar" for reporting the news of the landslide.

The Arunachal Times

STATE NEWS READERS FORUM EDITORIAL RING SIDE VIEW MONDAY MUSING SPARK

State News

Portion of NH 13 near Pasighat swept away, landslide blocks Likabali-Basar road

August 14, 2021

Staff Reporter

ITANAGAR, 13 Aug: A portion of National Highway (NH) 13, connecting Pasighat (East Siang), Pangin (Siang) and Aalo (West Siang), was reportedly swept away by a landslide triggered by heavy rain on Friday.

Sources in Pasighat informed that nearly 70-80 metres of the highway were swept away near Sirki waterfall, 4 kms from Pasighat, towards Pangin and Aalo. No casualty has been reported.

The Indian EXPRESS

Thursday, February 10, 2022

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Home / North East India / 4 children killed in massive landslide in Mizoram

4 children killed in massive landslide in Mizoram

An Assam-type building, in which a seven-member family lived, was swept away by the retaining wall that collapsed due to a landslip, triggered by rains, police said.

By: Express News Service | Updated: June 12, 2021 11:03:16 pm

LIVE BLOG

- Parliament Budget Session 2022 Live News:** Budget discussion to continue in RS, LS today 35 mins ago
- Mumbai News Live:** Prescribed uniforms should be followed in schools and colleges, Aadiya Thackeray says amid hijab row 37 mins ago
- Pune News Live:** 2,076 new Covid-19 cases; skies to turn overcast post noon 38 mins ago
- Coronavirus Omicron Live Updates:** India records 67,084 new cases, active count dips 40 mins ago



Rain-induced landslides are very common in urban areas in the hill-state of Mizoram because houses

The Sentinel

Thursday, 10 Feb 2022

HOME LIVE BLOG BREAKING NEWS TOP HEADLINES CITIES NE NEWS SENTINEL MEDIA SPORTS EDUCATION JOBS MORE

Meghalaya: RT-PCR Tests Not Required For Double Vaccinated Individuals Before Entering State If I Can Wear Cap & Go To Parliament, Why

Home / NE News / Arunachal News / Massive Landslide Hits ...

Massive Landslide Hits More Than 20 Locations In Arunachal Pradesh

Due to the massive landslides, it has severely affected movement of vehicle and people however no reports of deaths were reported.



Jobs in Assam

- 9 Feb 2022 4:02 PM: IGBRISH Assam Recruitment 2022: Assistant Professor / Statistician Vacancy, Job Openings
- 9 Feb 2022 4:01 PM: Assam Women's University Recruitment 2022: Multitasking Skilled Worker Vacancy, Job Openings
- 9 Feb 2022 4:00 PM: Kerditya Vidyalaya Misa Cantonment Recruitment 2022: Teacher, Job Openings
- 9 Feb 2022 4:00 PM: Tripura University Recruitment 2022: Programme Coordinator, DEO, Job Openings
- 9 Feb 2022 4:00 PM: CDAC Recruitment 2022: Project Manager, Project Leader, Job Openings

Jobs in Rest of NE

- 9 Feb 2022 4:00 PM: NIPOR Recruitment 2022: Junior Research Fellow / Student Internship Vacancy, Job Openings

By: Sentinel Digital Desk | 29 Sep 2021 3:36 PM

Sikkim Express 25/08/2021

Active landslides threaten to cut-off Daramdin constituency

KUSHHANGLEEWANG

GEYZING, August 24: Daramdin constituency in West Sikkim is at risk of being completely isolated from rest of the State as Soreng-Sombaria road is on the verge of sinking due to continuous landslips at Muneew Tareybhair stretch, about seven kilometres from Sombaria bazaar. Soreng-Sombaria road which acts as a lifeline for the people of Daramdin constituency is gradually sinking due to mud slips triggered by incessant rains. The route that links Daramdin constituency via Jorethang with other places is also blocked at Bojek slip since a long time and restoration work is underway. Dubbed as "sorrow of Daramdin", Bojek slip remains closed most often, especially during monsoon, as occurrence of landslides become rampant at that particular stretch of Bojek slip. The commuters and drivers have been facing hardships while plying through the dilapidated Muneew Tareybhair stretch. As the road at Muneew Tareybhair has become almost inaccessible, passengers and locals often have to pull the vehicle entangled in the middle by using ropes. "Plying through Muneew Tareybhair stretch has become troublesome for the drivers. It tests the patience and experience of a driver but we must drive through this dangerous stretch road despite obvious risk to us and passengers," said a driver. Soreng-Sombaria road serves as a road connectivity that links Daramdin constituency with urban areas such as Jorethang, Soreng, Dentam, Bermiok and Geyzing. In case Soreng-Sombaria road at Muneew Tareybhair stretch collapses, Daramdin constituency will remain completely cut-off from other constituencies. Considering the vulnerability and inevitable collapse of the stretch of Muneew Tareybhair road, it seems that the concerned department must chalk out a concrete plan in advance for restoration of the road as if the road collapses it may take time to repair the road network established at a steep and sinking stretch. The people of Daramdin constituency have requested the concerned department to look into the matter immediately. The road stretch of Muneew Tareybhair needs an urgent attention from authorities, said a local.



Guwahati News

Landslide In Guwahati's Borbari, 1 Labourer Dead



Protdin Bureau

Published on: 06 Aug, 2021, 5:47 pm

ANNEXURE-III: ONLINE MEDIA



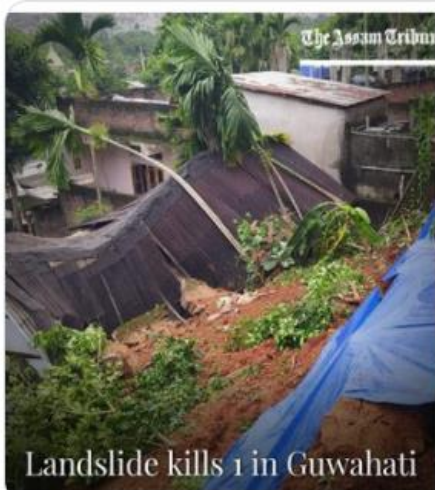


The Assam Tribune
@assamtribuneoff

A landslide that occurred in the wee hours at Water Works Colony, Pandu on Friday killed a 73-year-old woman and injured three others. The injured were later admitted to the hospital for treatment.

Photo: Subhamoy Bhattacharjee

#TheAssamTribune #GuwahatiLandslide



2:25 pm · 18 Jun 2021 · Twitter Web App

DownToEarth

CLIMATE CHANGE

After dry spell, heavy rains damage trans-Arunachal highway

Despite a rain deficit, connectivity is affected due to heavy landslides on critical roads connecting the remote Himalayan state with the rest of the world

By Anupam Chakravarty
Published: Monday 16 August 2021



Home Itanagar Arunachal Crime North East National International Sports Fest

9411 8122884208 www.internationaltribune.org

Itanagar- Part of NH-415 collapses after heavy rain

Incassant rain leaves trail of destruction in state capital

May 31, 2021

0 2 minutes read

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Home > Arunachal > Landslides play truant in Arunachal Pradesh; road connectivity badly hampered

Landslides play truant in Arunachal Pradesh; road connectivity badly hampered

By INSIDENE On Jun 30, 2021



EastMojo

Nagaland, News

Nagaland: Rockslide at NH-29; debris cleared for traffic

Two weeks after a massive rockslide along the stretch was reported, another rockslide occurred on Wednesday morning

By Mohit Kumar Kohli
February 17, 2021



ANNEXURE-IV: INVENTORY ATTRIBUTES

Slide ID	Unique Identifier for Landslide
Source	Newspaper/Media, SDMA, GSI or From Field
Slide No	Serial number
State	State in NER where landslide occurred
District	District where landslide occurred
Name	Name of the event
Location	Specific location from SDMA reports. General location – neighbourhood, village name etc from newspapers
NH/SH Cutoff/Affected	If National Highway, State Highway or any other important road was cutoff or affected
Latitude	Exact Latitude from GSI/SDMA reports General Latitude from Newspaper Articles
Longitude	Exact Longitude from GSI/SDMA reports General Longitude from Newspaper Articles
Date of Event	Date of the event
Time of Event	Time of event wherever possible
Type of Event	Type 1, 2 or 3 (explained in the next section)
Casualty	Casualty figures
Rainfall (24 hrs)	24 hours rainfall prior to the event

ANNEXURE-V: CODE SAMPLE

```
*C:\Users\NESAC\Desktop\2020\python\gee_code.py - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

gee_code.py
1 import ee
2 from ee_plugin import Map
3
4 #Code for displaying Pre and Post Imagery
5
6 # Processing polygon_ls
7 polygon_ls = "users/tusharmontfort/senapati"
8
9 aoi = ee.FeatureCollection(polygon_ls);
10
11 # ENTER THE TIME FRAME for the first image
12 start_date1 = '2020-10-01'
13 end_date1 = '2020-10-23'
14
15
16
17 # Import the Sentinel 2 collection as sentinel 2
18 sentinel2_1 = ee.ImageCollection("COPERNICUS/S2")
19
20 # Import the boundary
21 ls_bnd = ee.FeatureCollection(polygon_ls)
22
23 # Filter the image collection using filterBounds() and filterDate() method.
24 # Sort the collection by cloud cover metadata
25 image1 = sentinel2_1 \
26     .filterBounds(ls_bnd) \
27     .filterDate(start_date1, end_date1) \
28     .sort('CLOUDY_PIXEL_PERCENTAGE', False) \
29     .map(lambda image1: image1.addBands(image1.metadata('system:time_start')))\
30     .mosaic()\
31     .clip(ls_bnd)
32
33
34 #TRUE COLOUR COMPOSITE
35 visualization_TCC = {
36     'min': 0.0,
37     'max': 3000,
38     'bands': ['B4', 'B3', 'B2']
39 };
40
41 # FALSE COLOUR COMPOSITE
42 visualization_FCC = {
43     'min': 0.0,
44     'max': 3000,
45     'bands': ['B8', 'B4', 'B3']
46 };
47
48 LS_Scene = ee.Image(image1).select(['B4', 'B3', 'B2', 'B8']);
49
50 Map.addLayer(LS_Scene, visualization_TCC, 'true_color_composite_pre');
51 Map.addLayer(LS_Scene, visualization_FCC, 'False_color_composite_pre');
```

NERDRR TEAM FOR LANDSLIDE

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